

Network Devices -- Guided Notes

Unit tp-2.7 | CompTIA Network+ N10-009 Obj. FC0-U71 2.7 | N10-008 FC0-U71 2.7

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

A _____ operates at Layer 2 (Data Link) and forwards traffic based on _____ addresses. It creates a _____ collision domain per port, isolating each connected device. This is the primary device used to connect computers within a local area network (LAN).

Question 2

A _____ operates at Layer 3 (Network) and forwards traffic between different networks based on _____ addresses. The router in a home connects the local network to the _____ (Internet). It is the default _____ for devices on the LAN.

Question 3

A WAP (Wireless Access Point) provides _____ network connectivity by broadcasting an _____ using a radio signal. A home wireless router combines a router, switch, and _____ into one device. An enterprise may have _____ WAPs managed by a central wireless controller.

Question 4

A _____ (cable/DSL modem) converts the signal from an ISP (Internet Service Provider) into a format the local network can use. _____ modems use a coaxial cable connection. _____ modems use a telephone line. The word modem comes from _____ and _____ because it converts between analog and digital signals.

Question 5

A _____ is a network security device that monitors and controls incoming and outgoing traffic based on _____ rules. A hardware firewall is a dedicated _____ device protecting the entire network. A software firewall runs on an _____ device and protects only that machine.

Question 6

A _____ (network hub) is a legacy Layer 1 device that broadcasts every packet to _____ ports. This creates a _____ collision domain and wastes bandwidth. Hubs have been replaced by _____ in all modern networks because switches forward traffic only to the intended recipient.

Question 7

PoE (Power over Ethernet) allows a _____ cable to carry both data and _____ power. This is used to power devices such as _____, _____, and _____ (list at least two) without requiring a separate power outlet at each device location.

Question 8 -- Real World Application

REAL WORLD: A school IT team is upgrading a computer lab with 30 student computers. They currently have one old hub connecting all machines. They notice that when multiple students download files simultaneously, everyone's connection slows to a crawl. Explain why this is happening and how replacing the hub with a managed switch would solve the problem.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/tp-2-7-network-devices/tp-2-7-network-devices-answer-key.pdf
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